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Engineering Innovation Challenge 2022

Med IT Ez

Team U-2

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Project Abstract

The increasing population of elderly in Singapore calls for a change in our healthcare system to better accommodate the older generation. However, the healthcare sector faces challenges in recruiting health workers to meet the manpower demand. Therefore, innovative solutions such as autonomous machines should be considered to cater for better healthcare for the elderly.

The prototype of our final project "Med IT EZ" is built and tested. A down-scaled version of "Med IT EZ", which is a single cell of the cabinet, was built to imitate the full functionality of the final project. In this report, the process of prototype development will be discussed. The materials and components that are used by the prototype and the purpose they serve are also listed down. Next, a review on how the prototype collects and collates the data, and what actions are to be carried out based on the collected data is given. Some special features of the prototype such as human or object differentiation and face recognition are also included in this report. Finally, after testing out the feasibility of the prototype, some limitations of the prototype were listed out so that major improvements can be added to the final project. The risks of using "Med IT EZ" are also discussed and potential ways to alleviate or remove the risks and hazards were also provided.

Project Report

1. Background

With increasing life expectancy and low fertility rates, 25 percent of the population in Singapore is predicted to reach the age of 65 and older by 2030, dominating a quarter of the whole population of Singapore. (Lin, 2021) This rapid shifting in demographics would require the healthcare system to be prepared to accommodate the daily needs of our older generation. In today's world, the older generation faces challenges and problems such as immobility and forgetfulness which will definitely increase the difficulty of completing everyday tasks.

Based on the Singapore Census of Population, the number of residents in resident households aged 65 years and over who had mobility issues grew from around 25,500 to 50,000 in a mere twenty years, i.e. from 2000 to 2020. (Ministry of Health, Singapore, 2021) Diseases like Alzheimer's and dementia also cause elderlies to forget daily life routines, such as taking medicine and buying groceries. According to research from the Ministry of Health, Singapore, 1 in 10 seniors who are aged 60 and above in Singapore face challenges such as dementia. This number is expected to rise in the future with increased life expectancy.

On top of these complications, most elderlies often experience neglect, sometimes even by their own children, as the children have their own obligations and responsibilities too. Sadly, although life expectancy has increased, the quality of life seems to have stagnated throughout the past few years.

Therefore, new solutions should be developed to help elderlies lead an independent and healthy life during old age. Increasing the number of healthcare workers or caregivers seems like a logical remedy. However, the healthcare sector faces severe challenges in recruiting enough workers to fulfill the manpower needs; other more innovative ways of catering to elderlies' healthcare needs, such as autonomous machines, should be considered for more holistic healthcare for the elderly.

2. Purpose

"Med IT Ez" prototype, a single cell version of the final product equipped with essential functions is developed. The prototype aims to create a timetable for taking medication, avoid the machine from being wrongly used and send emails to third parties instantly when abnormal activities have been detected.

3. Prototype Design and Concept

3.1 Main Idea:

Implementing a Raspberry Pi system, human detection system, and machine learning to provide real-time data and transform unstructured data into structured data. This prototype serves to validate the functions of the final product and identify potential issues faced by users when using the product, such as unknown people appearing in the household, users appearing rarely or too often in front of the camera or when the drawer is opened at a wrong timing.

3.2 Materials:

- Plastic Drawer: 34cm (L) x 25cm (W) x 12cm (H) *1
- Magnetic Reed Switch: Arduino magnetic contact switch door sensor MC-38 *1
- Raspberry Pi 4 Model B 8GB RAM *1
- Buzzer *1
- LED *1
- Button *1
- Breadboard *1
- Cable - wiring
- Micro SD card 32GB *1
- Raspberry Pi 8MP Camera Module V2.1 *1

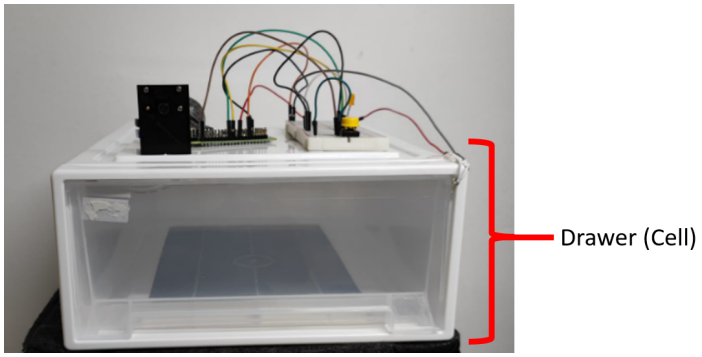


Figure 1. Overview of prototype

For the prototype, we have created a drawer representing a single cell of the final product.

3.3 Design

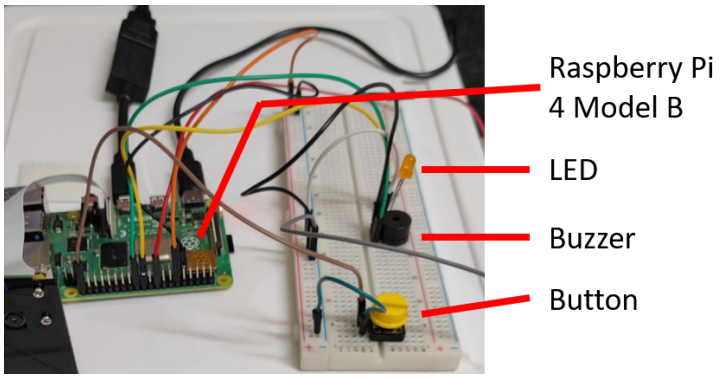


Figure 2. Basic Raspberry Pi Circuit

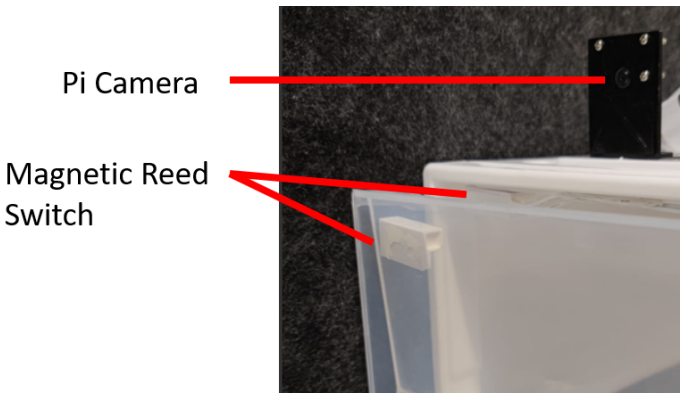


Figure 3. Placement of other components

3.4 Main components:

1. **Buzzers:** Making different sounds, ringing sound as an alarm to remind the elderly to take medicines, beeping sound to remind them to put back the medicine and emergency siren sound for emergency cases (eg. calling for help from nearby).
2. **Pi Camera:** Detect and track the activities of the elderly & allow guardians to monitor the elderly through the app.
3. **Cell:** Store medicine
4. **LED:** LEDs that correspond to the medicines to be taken will light up.
5. **Magnetic reed switch**: Detect if the cabinet is open

4. Procedures

1. The buzzer of the machine will signal the user with sounds when it is time to take medicine, according to the preset time by the user.
2. Each cell has its own LED light. The corresponding LEDs will light up to indicate which cells the user should take their medicine from.
3. Users will need to push the button to mute the buzzer after taking their medication.
4. When the medicine has been returned into the drawer, the LED will turn off.
5. If the user opens the cell at the wrong time, the buzzer and LED will switch on until the user closes the drawer, and an email is sent to the caregivers.
6. The Pi Camera does real-time face-recognition and motion tracking to detect abnormalities.
7. When abnormalities are detected, an email is sent to the caregivers.

5. Methods

The main program is implemented by coding in Thonny, a Python IDE downloaded in the Raspberry Pi 4 Model B. The code is programmed in several parts:

1. Magnetic reed switch sensor:
    a. Detects if the cell is opened or closed
    b. Activates buzzer and LED, sends email if the drawer is opened at the non-scheduled time
    c. Exports open/close data into a csv file
2. Medication time: user is allowed to input scheduled time to activate buzzer and LED
3. Pi camera:
    a. Human detection (Shown in Figure 4)

b. Facial recognition: the program is trained with several pictures of the user to identify them. (Shown in Figure 5)

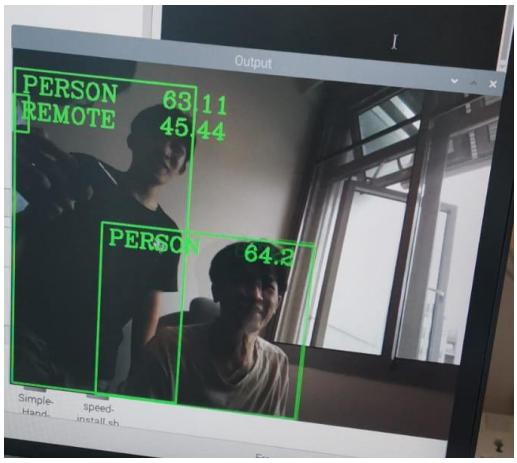


Figure 4. Image of Human Detection System



Figure 5. Image of Facial Recognition System

4. The assembly of prototype:
    a. The main circuit is assembled throughout the process of developing the program code
    b. The camera is directly connected to the Raspberry Pi via the MIPI CSI Port
    c. The remaining components are connected using wire and a breadboard
    d. Lastly, it is attached onto the drawer with adhesives

6. Limitations

The limitations of the prototype include:

- Unintentional trigger of the alarm when the door is still closed as the drawer is made out of plastic which is easily deformed affecting the accuracy of the switch door sensor.
- Different buzzer sounds might not be able to be differentiated by the elderly.
- Laggy camera which results in inaccurate face recognition.
- The process time between two actions is relatively longer due to the limited network and processor. For example, consecutively opening and closing the drawer in a short time will not activate the buzzer as the system is still processing the task of sending email.
- The prototype could be easily disassembled or destroyed since the Raspberry Pi system and circuit is fragile and exposed.
- Prototype is a single cell product therefore the feature of LED to indicate which cell to retrieve medicine cannot be tested.

7. Cost

Items	Quantity	Price (SGD)
Plastic Drawer: 34cm (L) x 25cm (W) x 12cm (H)	1	7.48
Arduino magnetic contact switch door sensor MC-38	2	3.38
Raspberry Pi 4 Model B 8GB RAM	1	253.20
Raspberry Pi 8MP Camera Module V2.1	1	55.00
Button	1	5
LED	1	5
Breadboard	1	5
Micro SD card 32GB	1	10
Micro-HDMI cable	1	15
Buzzer	1	5
	Total	364.06

8. Risk and Safety

1. Mechanical Hazard: The doors of the cells may cause injuries if body parts are in the way when it is being closed.
   ➔ Soft pads can be added to the edges of the cell to cushion body parts that are caught in the drawer.
   ➔ Soft closers can be added to slow down the closing speed of the door.
2. Electrical Hazard: One might get shocked or burned when coming into contact with the live parts of the circuit.
   ➔ All wirings of the Med IT Ez prototype ought to be well insulated and hidden from the user to avoid incidents of electric shock.
   ➔ The button is covered with a flexible plastic shield to avoid shocks if users' hands are wet.

9. Conclusion

"Med IT Ez " provides a solution to secure the wellbeing of elderlies. The prototype is able to organize the routine of taking medicine, and notify third parties such as family members or caregivers of the elderly after abnormal activities have been detected. The utilization of Raspberry Pi allows us to implement motion detection and tracking, human detection system and machine learning to transform unstructured data into structured data. This project has the potential to include more features and upgrades to be a smarter and safer machine which brings convenience to the aging population and shape a more inclusive and sustainable society by fulfilling the current needs of elderlies.

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